

03.08.2026

PRACTICAL-4

SETTING UP A LOCAL AREA NETWORK (LAN)

AIM:

To design, set up and configure a LAN connecting 4 computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

EQUIPMENT REQUIRED:

S.No	Equipment	Quantity	Specification
1.	Personal Computers (PC)	4	Ethernet (NIC) enabled.
2.	Network Switch	1	8/16/24 port, unmanaged/managed.
3.	Ethernet cables (Cat5e/Cat6)	4	Straight-through, RJ-45 connectors.

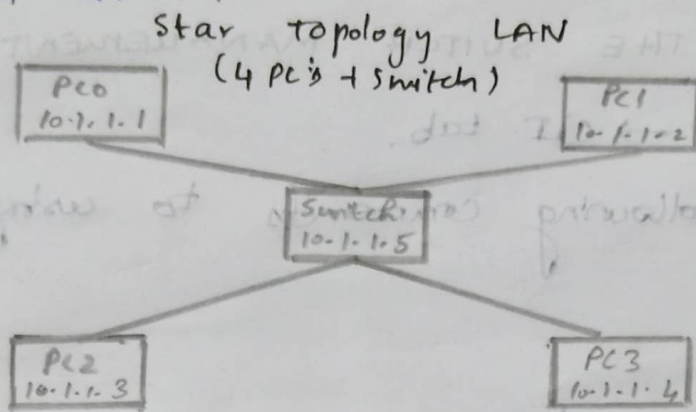
THEORY:

A LAN is a network that connects computers & devices within a limited geographical area such as a home, office, or a laboratory, allowing them to share resources such as files, printers and internet connections.

A star topology is used in this exp, where all PCs are individually connected to a central network switch. The switch operates at layer 2 of the OSI model and forwards data frames between connected devices based on their MAC addresses. Every device on a LAN must be assigned an IP address within the same network range to be able to communicate with one another.

NETWORK DIAGRAM:

The LAN was implemented using a Star Topology as shown below, with 4 PC's connected individually to a central switch.



IP Address Configuration Table:

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

IMPLEMENTATION IN CISCO PACKET TRACER:

Since physical hardware is limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software, following the steps below.

STEP 1: PLACE THE DEVICES

- 1.) Open Cisco Packet Tracer
- 2.) From the device category panel, select End Devices and drag 4 PCs onto the workspace
- 3.) Select switches and drag one switch (e.g. 2960 series) onto the workspace.

STEP 2: CABLE THE DEVICES

- 4.) Click connections → select copper Straight-Through cable.
- 5.) Click each PC, then click the switch, select an available Fast Ethernet port to complete each connection.
- 6.) Repeat for all four PC's (ports Fa0/1 to Fa0/4). Green link lights confirm a successful physical connection.

STEP 3: ASSIGN IP ADDRESSES TO PCs

- 7.) Click a PC → Desktop tab → IP configuration.
- 8.) Enter the Static IP Address and Subnet Mask as per the IP table (Section 5) for each PC.

STEP 4: CONFIGURE THE SWITCH MANAGEMENT IP (OPTIONAL)

- 9.) Click the switch → CLI tab.
- 10.) Enter the following commands to assign a management IP:

enable

configure terminal

interface vlan 1

ip address 10.1.1.5 255.0.0.0

no shutdown

exit

STEP 5: VERIFY CONNECTIVITY USING PING

- 11.) Click a PC → Desktop tab → Command Prompt.
- 12.) Use the ping command to test the connectivity to the other PCs e.g., ping 10.1.1.2.
- 13.) Successful "Reply from..." messages confirm connectivity.

STEP 6: DEMONSTRATE DATA SHARING (SERVER SIMULATION)

- 14.) Add a server device (End Devices - Server-PT) and connect it to the switch.

15.) Assign it an IP address (e.g., 10.1.1.10, subnet mask 255.0.0.0) via its Desktop → IP configuration tab.

- 16.) Open the server's Services tab and enable the HTTP service.

- 17.) From another PC, open the Web browser (Desktop tab) and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

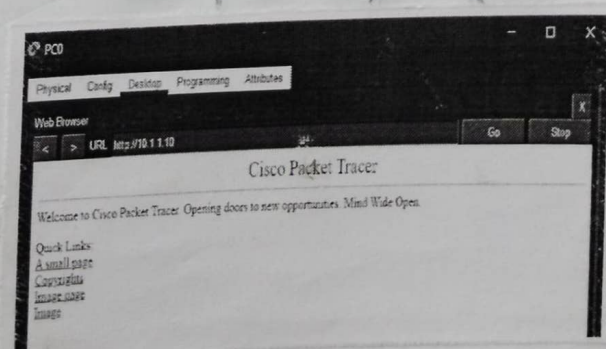
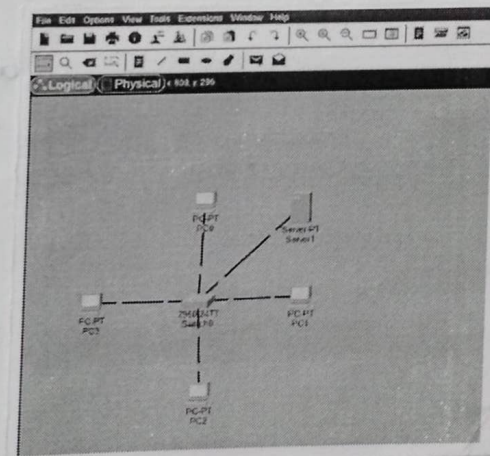
STEP 7: OBSERVE PACKET FLOW IN SIMULATION MODE

18.) Switch to simulation mode (bottom right of the workspace).

19.) Send a ping or web request between 2 devices and use the Play/capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.

OBSERVATION:-

All 4 PC's were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet (10.0.0.0/255.0.0.0). Ping tests between all devices returned successfully replies, confirming end-to-end connectivity. The shared folder/Server resource was successfully accessed from other computers on the network, confirming that the LAN was functioning correctly for resource sharing.



RESULT:

The LAN was successfully designed and configured. Connectivity between all PC's was verified using the ping command, and resource sharing was successfully demonstrated.